



“University for Industry”

Addis Ababa Science and Technology University

Office of Director for Graduate Programs

Guideline for Submission and Approval of Graduate Research Proposal, Thesis and Dissertation

December 2018

Addis Ababa, Ethiopia

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List of Abbreviations and Acronyms

AC	Academic Commission
BoE	Board of Examiners
CAS	College of Applied Sciences
CEME	College of Electrical and Mechanical Engineering
DGC	Departmental Graduate Committee
DGP	Director for Graduate Programs

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1. Introduction

This document has been prepared to provide guidelines for writing graduate research proposals, Master's thesis and PhD dissertation at Addis Ababa Science and Technology University (AASTU). The document also aims in providing the necessary information regarding integrity in research, thesis or dissertation submission and approval procedures.

It is the student's responsibility to read and fulfill the requirements presented here and to submit a document of high quality. Proposals, Theses and Dissertations that do not fully comply with this document shall not be accepted. Correct grammar, punctuation, and spelling should always be used, and these aspects are the responsibility of the student. Students should consult their respective department and advisors for specific content requirements.

All students have the responsibility to conduct and report the results of their research in an open and ethical manner. A thesis/dissertation represents the culmination of years of academic preparation and uniquely expresses a student's training, skills, and ideas. A thesis/dissertation therefore deserves the student's greatest effort and, as with all scholarly work, demands compliance with the highest ethical standards. Ethical misconduct in this regard may lead to severe consequences including complete rejection of the thesis/dissertation.

Ethical misconduct in research is any "fabrication, falsification and plagiarism in proposing, conducting or reporting research or other scholarly activities". Fabricating research findings refers to making up results, and falsifying research results refers to altering, misrepresenting, or selectively reporting findings. Each of these acts violates the integrity of the research process and constitutes a serious breach of accepted ethical standards. Ethical research practice therefore requires that students at all stages of the research process conduct their research in an open and honest manner.

2. Preparation and Approval of Graduate Research Proposals

2.1. General Background

A graduate research proposal is an outline of proposed research work that is designed to:

- Define a clear question/hypothesis, set objectives and approach to answering it
- Explain how it adds to, develops (or challenges) existing literature in the field
- Highlight its originality and significance

Graduate students should begin investigating potential research areas and discussing the feasibility of possible research work with appropriate faculty at beginning their graduate careers. The ideal situation would be for a student to have chosen the general research area and to have received an indication from a faculty member that he or she would be willing to serve as advisor to the student before the end of the first semester.

As per the guideline and regulations of AASTU, a Master student must have an approved Thesis Proposal before the beginning of third semester and a PhD student a Dissertation Proposal by the end of the third semester.

Generally, a Master study must make a substantial work based on independent research which shows a sound knowledge of the subject of the research, evidence of the exercise of some independence of thought and the ability of expression in clear and concise language. The justification of the study will state how the proposed study is expected to fulfill these requirements. A PhD study must make a substantial and original contribution to knowledge, such as the discovery of new knowledge, the formulation of theories, or the innovative re-interpretation of known data and established ideas. The justification of the study will state how the proposed study will fulfill this requirement, i.e., a statement of the need for the proposed study in terms of its anticipated contribution to the field.

2.2. Procedure for Preparation and Approval of Graduate Research Proposals

The procedure for preparation and approval of the proposals is given below.

1. In consultation with his/her advisor(s), the student identifies a research topic and develops a proposal which addresses the following points: nature and significance of the problem, expected objectives and outcome, description

of the procedures used to solve the problem, timetable for the work and budget breakdown and etc.

2. Advisor(s) critically assess the Master's Thesis/PhD Dissertation proposal and check whether the student's proposal is up to standard and prepared following this guideline.
3. The student shall submit three hard copies and soft copy, signed by his/her advisor(s), to the department.
4. Following the DGC meeting on the submitted proposal, DGC chairperson passes a copy of the Master Thesis/PhD Dissertation proposal to two evaluators who are expert in the area and arranges proposal defense presentation.
5. Assigned evaluators shall thoroughly review the proposal and evaluate using **AASTU/GP-F9a** form for Masters and **AASTU/GP-F9b** form for PhD proposals, after which it may a) return the proposal to the student for clarification, modification, rewriting or rework, or b) approve the proposal as submitted.
6. Student makes corrections to accepted proposal and submit 4 signed copies of the corrected proposal (with signatures of evaluators and advisor(s) on the approval page of the proposal) to the department after advisor(s) checked that all agreed corrections and suggestions of assigned evaluators are included. Students with rework proposals should follow procedures 1-6 again.
7. After checking the inclusion of all corrections and suggestions, the DGC chairperson duly signs the approved copies of the proposal and sends to the college AC.
8. The college AC will assess the proposal in terms of the guidelines used and feasibility including the cost breakdown. The AC will return incomplete proposals to the respective department for improvement.

9. Following the AC decision, the College Associate Dean for GP approves the proposal by duly signing each copy. One copy of the proposal should be sent to the office of Director for Graduate Program (DGP).

10. Sponsored students request for research fund as per ***Thesis/Dissertation Research Grant Guideline***.

The above procedure is summarized in the flowchart given in the ***Appendix 4***.

NB: As per the guidelines and regulations of AASTU, if a student does not have an approved Master's Thesis Proposal before the beginning of third semester or a PhD Dissertation Proposal by the end of third semester, then it is taken as the student is not making "Normal Progress". The DGC and AC must be made aware of the reasons for the delay, expected timeline for completion, and a meeting may be requested with the student and advisor to discuss the circumstances.

2.3. Structure and Contents of Graduate Research Proposal

The research proposal should include the following sections as per the sequence below:

Preliminary Pages	• Title Page • Approval Page • Table of Contents • Abbreviation and Acronyms (if any) • List of Figures (if any) • List of Tables (if any)
Main Body of Proposal	1. Introduction • Problem statements • Significance of the study • Objective of the study 2. Related Literature Review 3. Research Methodology 4. Work Schedule and Cost Breakdown References Appendix (if any)

2.3.1. Title Page

- The Title Page is the first page of the proposal. It is considered page “i”, but the number does **not** appear on the page.
- The title page should include the following:
 - i. AASTU logo
 - ii. Full title of proposal;
 - iii. Full name of student;
 - iv. Full name of advisor(s);
 - v. Name of the Department (stream);
 - vi. Name of the College;
 - vii. Name of the university;
 - viii. Month and year of submission.
- If the title runs onto more than one line, use inverted pyramid style (first line longer than the second) and double spaced.
- Refer **Appendix 1** for the title page format specifications and sample title pages.

2.3.2. Approval Page

- The approval page is page ii.
- Refer **Appendix 1** for sample approval page.

2.3.3. Table of Contents

- The Table of Contents lists in sequence all relevant subdivisions of the proposal with their corresponding page numbers.

It includes:

Approval Page

List of Tables (if any)

List of Figures (if any)

Abbreviation and Acronyms (if any)

1. Introduction

- Problem statements
- Significance of the study

- Objective of the study
- 2. Related Literature Review
- 3. Research Methodology
- 4. Work Schedule and Cost Breakdown

References

Appendices (if any)

- Generate the table of contents at the very end of your proposal work, since page numbers often change due to last-minute additions/deletions.

2.3.4. List of Tables

- The list shows the exact titles or captions of all tables in the text and appendices, together with the page number of each table, and must be listed in sequence.
- If the whole proposal contains only one or two tables, then a List of Tables is not necessary.
- A period shall be placed after the number, before the caption (e.g., Table 1. Caption of table).
- Single space individual captions/titles that run over onto two or more lines.

2.3.5. List of Figures

- Figures include graphs, maps, charts, engineering drawings, photographs (plates), sketches, printed images, and any other form of illustration that is not a table.
- The exact titles or captions and their corresponding page numbers must be listed in sequence.
- Figures, including those in the appendices (if any), should be numbered consecutively throughout the proposal.
- If the whole proposal contains only one or two figures, then a List of Figures is not necessary.
- A period shall be placed after the number, before the caption (e.g., Fig. 1. Caption of figure).
- Single space individual captions/titles that run over onto two or more lines.

2.3.6. List of Abbreviations and Acronyms

- If abbreviations and acronyms are used in the proposal, they should be explained in Abbreviations and Acronyms page, even though the full names are given at first use.
- This list should be the last item in the preliminary section. It serves as a ready reference to readers not familiar with the abbreviations used in the proposal.
- Universally recognized scientific symbols (such as CO₂, cm, mm, kg, DNA, etc.) need not be listed.
- List of Illustrations, List of Equations, List of Special Symbols, Definitions, etc. should be included, where applicable, under the respective title.

2.3.7. Introduction

The introduction gives an overview of the research project highlighting the significance of the research work that is proposed to be carried out. It explains the background of the project, focusing briefly on the major issues of its knowledge domain and clarifying why these issues are worthy of attention. In addition to the general background, the introduction should highlight the research gap by reviewing relevant literature. It then proceeds with the concise presentation of the research statement, which can take the form of a hypothesis, a research question, a project statement, or a goal statement. The research statement should capture both the essence of the project and its delimiting boundaries, and should be followed by a clarification of the expected outcome and contribution to the existing knowledge.

The introduction should endeavor, from the very beginning, to catch the reader's interest and should be written in a style that can be understood easily by any reader with a general science background. It should cite all relevant references pertaining to the major issues described, and it should close with a brief description of each one of the sections that follow.

2.3.8. Related Literature Review

The literature review should be a critical analysis of the existing knowledge on research problems, objectives and methods. It includes strength of previous studies, limitations and the gaps. In addition, the literature review should give a detailed account of the theoretical background and related work. The review need to highlight the main

issues in the research area and the reviewer need to reconcile any conflicting hypothesis/results reported by different authors by providing his own interpretation and outlook on the matters of contention. A critical review of literature needs to be presented in an organized and synthesized manner with coherent flow of ideas. It clarifies, strengthens and directs each stage of research and indicates what is new in a proposed study. Furthermore, the literature review should give a comprehensive coverage of the state of the art in the field of study and highlight the research gaps that need further investigation. It is important to note that the review should focus on more relevant and recent literature.

2.3.9. Research Methodology

Research methodology is the procedures or techniques used to identify, select, process, and analyze data about the proposed research project. The research methodology section should answer the following main questions: How will the data be generated? How would it be analyzed? Where and under what conditions will the study be undertaken? What materials and measuring instruments will be used? What statistical tools will be used for uncertainty analysis and validation? What mathematical and numerical tools will be used?

This section should give clear, specific, appropriate and credible procedures that will be followed to attain the proposed objectives of the study. The research design planned for use should be clearly stated. The research methodology section possibly contains the following information: definition of terms and variables, description of the study area or experimental site, schematics of experimental setup, study design or experimental design, description of population (for relevant discipline), sampling methods, data collection and analysis techniques, method of uncertainty analysis, statistical analysis, etc.

2.3.10. Work Schedule and Cost Breakdown

The work plan refers to budgeting of time for the implementation of the research project. This is to assign dates for the completion of various activities of the proposed research. The act of submitting the work plan in the research proposal systematizes the study and minimizes the natural tendency to procrastinate. A detailed work plan showing jobs to be done in the main phases of the study and the time sequence (the

operating schedule), more of it conveniently presented in the form of Gantt chart, is essential, if the project is to be carried out smoothly and efficiently. A detailed budget breakdown for each and every expense of the project should also be presented in the form of table in accordance to the timeline of project activities.

2.3.11. References

- All references cited in the body text of the proposal should be listed in the reference section and vice versa.
- The list of references should appear as a consolidated list with references listed either alphabetically or sequentially as they appear in the text of the proposal, depending on the reference style used. Refer **section 4.5** of this document for the details reference styles and formats.
- Spacing and font size should be consistent inside a single reference, and there should be single spacing between two different references.

3. Preparation and Approval of Thesis and Dissertation

The terms Thesis and Dissertation refer to written reports of research studies undertaken in partial fulfillment for the award of Master and Doctor of Philosophy (PhD) degrees, respectively. These documents, thesis and dissertation, should demonstrate a mastery of a body of knowledge on the subject. They represent original scholarly research and writing, and their content and style make statement about the student, the advisors who have guided the student, and the AASTU. Hereinafter, both the thesis and the dissertation shall be referred to by the term “thesis” unless the information is specific to Master’s or doctoral students. Where format requirements differ, the specific requirements for each document are indicated.

3.1. Procedure for Submission and Approval of Thesis

The procedure for submission and approval of the thesis is given below.

1. Student commences writing the thesis by utilizing this document for format and style requirements
2. Advisor(s) check whether the document reaches the final stage.
3. The student shall submit three copies, signed by his/her advisor(s), to the department by completing **AASTU/GP-F5a** form for Master’s thesis and **AASTU/GP-F6a** form for PhD dissertation. A Master’s student has to provide evidence and copy of article submitted to peer review journal and a PhD student has to submit the copies of published articles along with the dissertation.
4. DGC shall evaluate whether the submitted article is publishable (for Master’s thesis) and the thesis/dissertation is defensible? If the DGC decides the thesis/dissertation is defensible, it nominates the Board of Examiners (BoE) and recommend to College AC.
5. AC approves the Board of Examiners.
6. DGC schedules the Thesis/Dissertation defense date and send a copy of the document to each evaluator before one month for Master thesis and four month for PhD dissertation.

7. Assigned examiner will thoroughly review the thesis/dissertation and evaluate using **AASTU/GP-F5b** form for Master's thesis and **AASTU/GP-F6b** form for PhD dissertation and the chairperson shall use **AASTU/GP-F5c** form for Master's thesis and **AASTU/GP-F6c** form for PhD dissertation.
8. If the thesis/dissertation is accepted and requires correction, student makes corrections to accepted thesis/dissertation and submit 4 signed hardbound copies of the corrected final thesis/dissertation (with signatures of examiners and advisor(s) on the certificate page and declaration page of the thesis/dissertation) to department after the BoE checked that all agreed corrections and suggestions of assigned examiners are included. Students with rejected thesis/dissertation should follow DGC's and BoE's decisions.
9. After checking the inclusion of all corrections and suggestions, DGC chairperson duly signs the approved copies of the thesis/dissertation and sends to the college AC.
10. The college AC shall approve the thesis/dissertation after checking the guidelines used and procedure followed. The AC will decline incomplete thesis/dissertation to the respective department for improvement.
11. Following the AC decision, the college shall send one copy of the hardbound copy to the office of DGP.

The above procedure is summarized in the flowchart given in the **Appendix 5**.

3.2. Contents of Thesis

The thesis should include the following sections and sequence:

Preliminary Pages	Cover Page Title Page Approval Page Declaration Dedication (optional) Abstract Acknowledgements Table of Contents Abbreviations and Acronyms (if any) Lists of Tables (if any) List of Figures (if any)
Text of Thesis	Chapter 1 Introduction Chapter 2 Literature Review Chapter 3 Methodology Chapter 4 Results and Discussion (Students may write their results and discussion in separate chapters.) Chapter 5 Conclusions and Recommendations References Appendices List of Publication

3.2.1. Cover Page

- The cover page is the first page of a thesis and it is the only page of a thesis for which a page number is not assigned.
- It is printed on **hard paper and required only during final submission after defense.**
- The following are requirements for the cover page (refer to ***Appendix 2 and 3*** for more details).

- Thesis Spine

The spine must be entirely lettered in gold, using a 18-point font and must contain the following:

- i. Name of student;
- ii. Title of the thesis
- iii. AASTU YYYY

- Cover Page

- The front cover must be **BLUE WITH WHITE FRAME** for Doctor of Philosophy Dissertation and **BLUE WITHOUT FRAME** for Master's Thesis.
- The front cover must be entirely lettered in gold using 18-point white block font and contain the following:
 - i. AASTU Logo;
 - ii. Name of the University;
 - iii. Title of thesis;
 - iv. A Master's Thesis/ A Doctoral Dissertation;
 - v. by
 - vi. Name of student;
 - vii. Name of the department;
 - viii. Name of the college and
 - ix. Year of submission

3.2.2. Title Page

- The Title Page is the first page of the thesis proposal. It is considered page “i”, but the number does **not** appear on the page.

- The title page should include the following:
 - AASTU logo
 - Name of the university;
 - Full title of thesis;
 - Full name of student;
 - Degree for which the thesis is submitted;
 - Name of the department;
 - Name of the college;
 - MONTH and YEAR of submission.
- If the title runs onto more than one line, use inverted pyramid style (first line longer than the second) and double spaced.
- Refer **Appendix 2** for the title page format specifications and sample title pages of thesis.

3.2.3. Approval Page

- Refer **Appendix 2** for sample approval pages.

3.2.4. Declaration Page

- Refer **Appendix 2** for sample declaration pages.

3.2.5. Dedication (optional)

- The Dedication page is optional and can be deleted if the student decides not to use it.

3.2.6. Abstract

- The abstract is a digest of the entire thesis and it must bring out the salient points about your work, highlighting your achievements. It should **not** include any references. Text should be double spaced.
- Abbreviations or acronyms must be preceded by the full terms at the first use.
- It includes a brief statement of the problem and objectives of the study, a concise description of the research method and design. Summary of the major findings including their significance and conclusions. Abstract should end with at most “five key words” of the study.

3.2.7. Acknowledgments

- This page is for the author to express professional and/or personal indebtedness, people or institutions that have contributed to the content of his/her thesis, permission(s) to use previously copyrighted materials, receipt of grant funds, recognition of “readers”, etc. is acknowledged on this page.
- Students must be consistent with the use of the third ("the author") or first person throughout.
- This page is optional, but if included, it is paginated (lower case Roman numerals) and listed in the Table of Contents. If this page is included, head the section as “Acknowledgments”.

3.2.8. Table of Contents

- The Table of Contents lists in sequence all relevant subdivisions of the thesis with their corresponding page numbers.

It includes:

Approval

Declaration

Abstract

Acknowledgements

List of Abbreviation and Acronyms

List of Tables

List of Figures

List of Illustrations, List of Equations, List of Special Symbols, Definitions, etc. (where applicable)

Text of the Thesis

References

Appendices

List of Publications

3.2.9. List of Abbreviations and Acronyms

See **section 2.3.6** of this document for more details.

3.2.10. List of Tables

See **section 2.3.4** of this document for more details.

3.2.11. List of Figures

See **section 2.3.5** of this document for more details.

3.2.12. Text of Thesis

- The body of a thesis proposal normally consists of sections which are organized as chapters.
- A chapter may be divided into major sections and subsections.
- Main or primary headings within chapters are to be centered while sub-headings are left justified.
- Quaternary headings are **not** listed in the Table of Contents.
- The main sections and subsections of a chapter may be identified by numbers where the former are regarded as being the first level. For example, Sections 2.1 and 2.2 would denote two consecutive main sections in Chapter 2, and Sections 3.1 and 3.2 would denote two consecutive main sections in Chapter 3.
- A subsection would be found in a major section of a chapter, and is regarded as the third level. It should be numbered 2.1.1., 2.1.2., etc. The numbering style should be consistent throughout the thesis and should be limited to 4 levels.
- Students are advised to discuss the usage of tables and figures with their advisor before their inclusion in the thesis, as different disciplines have different preferences.
- Body of thesis is paginated with Arabic numbers, starting with number **1** on the first page of Chapter 1.
- Every page must be numbered consecutively.
- The Ph.D. student has two options in writing of his or her dissertation for the doctor of philosophy degree. One option is the traditional and more common format known as the “monograph” format. The other dissertation format is referred to the “manuscript” format.

- The monograph format will share a similar format with the traditional thesis. It will have sections including an introduction chapter, a literature review chapter, materials and methods chapter, results and discussion chapter (which may be written in two chapters), a summary, conclusions and recommendations chapter, a references chapter and an appendix.
- The manuscript format will first include a summary of the dissertation of perhaps twenty to thirty pages in length. In preparing the dissertation summary, the student will be cautious to minimize redundancy in making references to the respective manuscripts which follow in the dissertation while presenting the methods and presenting the results and discussion. This dissertation summary will then be followed by a collection of both published and unpublished articles on the subject matter that he or she and members of the advisory committee have written together. These articles will be either reprints of publications with prior permission from the journal editor or drafts in manuscript format. So, the manuscript format will have a summary of the dissertation which is followed by a collection of published and unpublished articles originating from the dissertation research.
- These two options are now offered to the doctoral students to provide some variety in the preparation of their dissertations. The two options also provide an opportunity for students who have published extensively to highlight their writing and publishing activities while working on their dissertations.
- These two formats should be discussed with the student's academic advisors.
- Decisions about the choice of format can be made during the preparation of the dissertation.

3.2.13. References

See **sections 2.3.11** and **4.5** of this document for more details.

3.2.14. Appendices

- Information or data that is too detailed for the main body of the thesis may be included as appendices.
- Appendices include original data, summary, sideline or preliminary tests, tabulations, tables that contain data of lesser importance, very lengthy quotations,

supporting decisions, forms and documents, computer printouts, detailed engineering drawings and other pertinent documents.

- Appendix materials should be grouped by type, e.g., Appendix A: Questionnaire, Appendix B: Original data, Appendix C: Tables of results.
- Appendices must be paginated consecutively with the main text.

3.2.15. List of Publications

- All publications (in journals and proceedings) that result from the study undertaken by the student while under supervision and during their candidature, and for which the student is the first or principal author, should be listed clearly and accurately.

4. General Style and Formatting

4.1. Paper Size and Margins

- A4 is the recommended thesis size.
- The left margin should be at least 1.5 Inch.
- The right, top and bottom margins at least 1 Inch.
- Margin specifications are meant to facilitate binding and trimming.

4.2. Font Type and Size

- The text of the proposal, including headings and page numbers, must be produced with the same font or typeface.
- The font size should be **12-point** and should not be scripted or italicized except for scientific names and terms in a different language.
- Bold print should be used for headings.
- Text in tables should not be less than 8-point.
- A font appropriate for a proposal is Times New Roman font.

4.3. Spacing

- The graduate research proposals, theses and dissertations should be 1.5.
- The following, however, should be single-spaced:
 - Quotations of three lines or more, indented and set in a block;
 - References or bibliography (except between entries);
 - Multi-line captions of tables and figures;
 - Appendices,
 - Headings or subheadings

4.4. Pagination

- All pages should be numbered consecutively throughout the graduate research proposals, theses and dissertations, including pages containing appendices and references.
- Page numbers should be centered either centrally or right flushed at bottom margins.
- Page numbers should appear by themselves and should not be placed in brackets, be hyphenated or be accompanied by decorative images.

- Text, tables and figures should be printed on one (1) side of each sheet only.
- Preliminary pages, except the title page, preceding 1. Introduction must be numbered in lowercase Roman numerals (ii, iii, iv, etc.).
- The title page should not be numbered although it is counted as page i.
- Page 1 is the first page of the 1. Introduction.

4.5. Reference Styles and Formats

- Students, in consultation with their advisor(s), are responsible for choosing a style of citation appropriate to the field and should follow the reference style and format consistently throughout the document.

Bibliography

The following documents were consulted and used in preparation of this guidebook.

1. Senate Legislation, *Addis Ababa Science and Technology University*, July 2017.
2. Thesis writing, examination and grading guidelines, *School of Graduate Studies, Addis Ababa University*, April 2012.
3. Research Proposal, Thesis and Dissertation Writing Manual *Second Edition*, *Haramay University*, October 2011.
4. Guidelines for Thesis Preparation, *Indian Institute of Technology Kanpur*.
5. Guide to thesis and Dissertation, *School of Graduate Studies, The University of Auckland*.

Appendix 1: Format Specifications and Sample Templates for Preliminary Pages of Proposals

Format Specification for Title Page for Proposals

Sample Title Page for Master's Thesis Proposal

Sample Title Page for Doctoral Dissertation Proposal

Approval Page



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

(Upper case, centered, bold, italic, 14-point font)

TITLE OF THE THESIS/DISSERTATION

RESEARCH PROPOSAL

(Upper case, centered, bold, 18-point font, double spaced)

<MSc/MBA Thesis or PhD Dissertation> Research Proposal

By

NAME OF STUDENT

(Upper case, centered, bold, 14-point font)

Advisor(s): NAME OF ADVISOR(S) (Upper case, centered, bold, 14-point font)

DEPARTMENT OF <INSERT NAME OF THE DEPARTMENT>

(<INSERT NAME OF STREAM>)

COLLEGE OF <INSERT NAME OF THE COLLEGE>

(Upper case, centered, bold, 14-point font, 1.5p spaced)

MONTH YEAR

(Upper case, centered, bold, 14-point font)



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

**ULTRA LOW ENERGY COMMUNICATION
PROTOCOL FOR IMPLANTABLE WIRELESS
BODY SENSOR NETWORK**

MSc Thesis Research Proposal

By

MENGISTU URGE KEBEDE

Advisor: MULATU GEBISA (Ph.D.)

**DEPARTMENT OF ELECTRICAL AND COMPUTER
ENGINEERING (COMMUNICATION STREAM)
COLLEGE OF ELECTRICAL AND MECHANICAL
ENGINEERING**

APRIL 2019



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

**THREE-DIMENSIONAL ANALYSIS OF
TUNNELLING EFFECTS ON STRUCTURES TO
DEVELOP DESIGN METHODS**

PhD Dissertation Research Proposal

By

MENGISTU URGE KEBEDE

Advisor: MULATU GEBISA (Ph.D.)

**DEPARTMENT OF CIVIL ENGINEERING
(STRUCTURAL ENGINEERING STREAM)
COLLEGE OF ARCHITECTURE AND CIVIL ENGINEERING**

APRIL 2019

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Appendix 2: Format Specifications and Sample Pages for Preliminary Pages of Master's Thesis

Spine and Cover Page Format Specifications for Master's Thesis

Title Page Format Specifications for Master's Thesis

Sample Title Page for Master's Thesis

Sample Declaration Page for Master's Thesis

Sample Approval Page for Master's Thesis

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Title of the thesis

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TITLE OF THE THESIS

(Lettered in gold, upper case, centered, bold, 18-point font, double spaced)

A MASTER'S THESIS

BY

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ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

**ULTRA LOW ENERGY COMMUNICATION
PROTOCOL FOR IMPLANTABLE WIRELESS
BODY SENSOR NETWORK**

By

MENGISTU URGE KEBEDE

A Thesis Submitted as a Partial Fulfillment to the Requirements for the Award of the

Degree of Master of Science in Electrical and Computer Engineering

(Communication Engineering)

to

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING

APRIL 2019

Declaration

I hereby declare that this thesis entitled “**Ultra Low Energy Communication Protocol for Implantable Wireless Body Sensor Network**” was prepared by me, with the guidance of my advisor. The work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted, in whole or in part, for any other degree or professional qualification. Parts of this work have been published in [state previous publication].

Author:

Signature, Date:

Witnessed by:

Name of student advisor:

Signature, Date:

Name of student co-advisor:

Signature, Date:

Approval Page

This is to certify that the thesis prepared by **Mr. Mengistu Urge Kebede** entitled “**Ultra Low Energy Communication Protocol for Implantable Wireless Body Sensor Network**” and submitted as a partial fulfillment for the award of the Degree of Master of Science in Electrical and Computer Engineering (Communication Engineering) complies with the regulations of the university and meets the accepted standards with respect to originality, content and quality.

Signed by Examining Board:

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Signature, Date:

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Signature, Date:

College Dean/Associate Dean for GP:

Signature, Date:

Appendix 3: Format Specifications and Sample Pages for Preliminary Pages Dissertation

Spine and Cover Page Format Specifications for PhD Dissertation

Title Page Format Specifications for PhD Dissertation

Sample Title Page for PhD Dissertation

Sample Declaration Page for PhD Dissertation

Sample Approval Page for PhD Dissertation

Name of Student



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A DOCTORAL DISSERTATION

BY

NAME OF STUDENT

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TITLE OF THE DISSERTATION

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NAME OF STUDENT

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A Dissertation Submitted as a Partial Fulfillment to the Requirements for the Award
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ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

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Author:

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Witnessed by:

Name of student advisor:

Signature, Date:

Name of student co-advisor:

Signature, Date:

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